Uncertainty-Aware Regime-Conditioned Reinforcement Learning for Portfolio Management

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Abstract

We propose UARC (Uncertainty-Aware Regime-Conditioned RL), a portfolio management framework that integrates three complementary components: a Bayesian Hidden Markov Model (HMM) for probabilistic market regime inference, an inverted Transformer encoder (iTransformer) for cross-asset feature extraction, and Implicit Quantile Networks (IQN) for distributional reinforcement learning. Unlike prior work that conditions portfolio agents on hard regime labels via argmax of the HMM posterior [1], UARC passes the full K-dimensional posterior distribution $p(z_t \mid x_{1:t})$ as an agent state signal, preserving uncertainty information during regime transitions. The iTransformer treats each asset as a token and runs self-attention across the asset dimension, capturing cross-asset correlations more directly than time-first attention. The IQN agent learns the full return distribution per portfolio allocation, enabling CVaR-penalized weight selection without a separate risk model. Evaluated on a five-asset portfolio (SPY, QQQ, TLT, GLD, SHY) over a four-year holdout period (2021–2024), all learned systems achieve an annualized Sharpe ratio of 0.576 versus 0.454 for equal-weight buy-and-hold (27% improvement). A hierarchical four-system ablation reveals a surprising finding: the iTransformer encoder implicitly subsumes HMM regime information, rendering explicit posterior conditioning redundant—an empirical result with important implications for hybrid probabilistic–RL architectures.

Keywords

Reinforcement Learning, Portfolio Management, Hidden Markov Models, Distributional RL, Market Regime Detection, iTransformer, CVaR, Implicit Quantile Networks

1 Introduction

Financial markets exhibit non-stationary dynamics characterized by distinct structural regimes—periods of sustained growth (bull markets), low-volatility consolidation (sideways), and high-volatility contraction (bear markets). Ang and Bekaert [2] demonstrated that regime-switching models significantly outperform single-regime alternatives in capturing the empirical distribution of asset returns, motivating portfolio management frameworks that explicitly condition allocation decisions on estimated market regimes.

Almahdi and Yang [1] introduced regime-conditioned reinforcement learning for portfolio management using an HMM to classify regimes and passing the argmax label to a DQN agent. While effective, this approach discards the full posterior distribution $p(z_t \mid x_{1:t})$ and reduces rich probabilistic information to a single integer. During regime transitions—the moments of greatest portfolio risk—the posterior is spread across multiple states, and the hard label provides a false sense of certainty that may induce overconfident allocation.

Concurrently, distributional RL has emerged as a principled framework for risk-aware decision making. Bellemare et al. [3] showed that learning the full return distribution improves both sample efficiency and final performance. Dabney et al. [4] extended this with Implicit Quantile Networks (IQN), which represent arbitrary return distributions via implicit quantile functions, enabling direct computation of risk measures such as CVaR without a separate risk model or parametric distributional assumption.

We propose UARC, which unifies these directions into a single architecture (fig. 1). Our contributions are:

(1) Passing the *full* Bayesian HMM posterior as state rather than argmax, preserving uncertainty during regime transitions.
(2) Employing an iTransformer encoder [8] that treats each asset as a token and runs attention across the asset dimension, capturing cross-asset correlations naturally.
(3) Using IQN [4] for distributional RL, enabling CVaR-penalized portfolio allocation directly from the learned return distribution.
(4) A comprehensive four-system hierarchical ablation on a four-year holdout (2021–2024) with a key negative finding: explicit regime conditioning does not improve over the no-regime IQN baseline, suggesting the iTransformer encoder implicitly captures regime structure from the 60-day lookback window.

2 Related Work

2.1 Regime-Switching Models

Hamilton [6] introduced the regime-switching framework for economic time series. Ang and Bekaert [2] demonstrated HMMs capture the asymmetry between bull and bear markets in international equity returns. Rabiner [10] provides the forward algorithm we use for causal posterior inference $p(z_t \mid x_{1:t})$ without lookahead—critical for avoiding temporal leakage.

2.2 Deep RL for Portfolio Management

Jiang et al. [7] introduced EIIE, establishing the continuous portfolio vector framework. Moody and Saffell [9] demonstrated direct reinforcement for trading rule optimization. Almahdi and Yang [1] is the closest prior work: HMM + DQN with hard regime labels. UARC extends this by passing the full posterior and replacing DQN with distributional RL.

2.3 Distributional Reinforcement Learning

Bellemare et al. [3] introduced C51; Dabney et al. [5] proposed QR-DQN; Dabney et al. [4] introduced IQN with implicit quantile

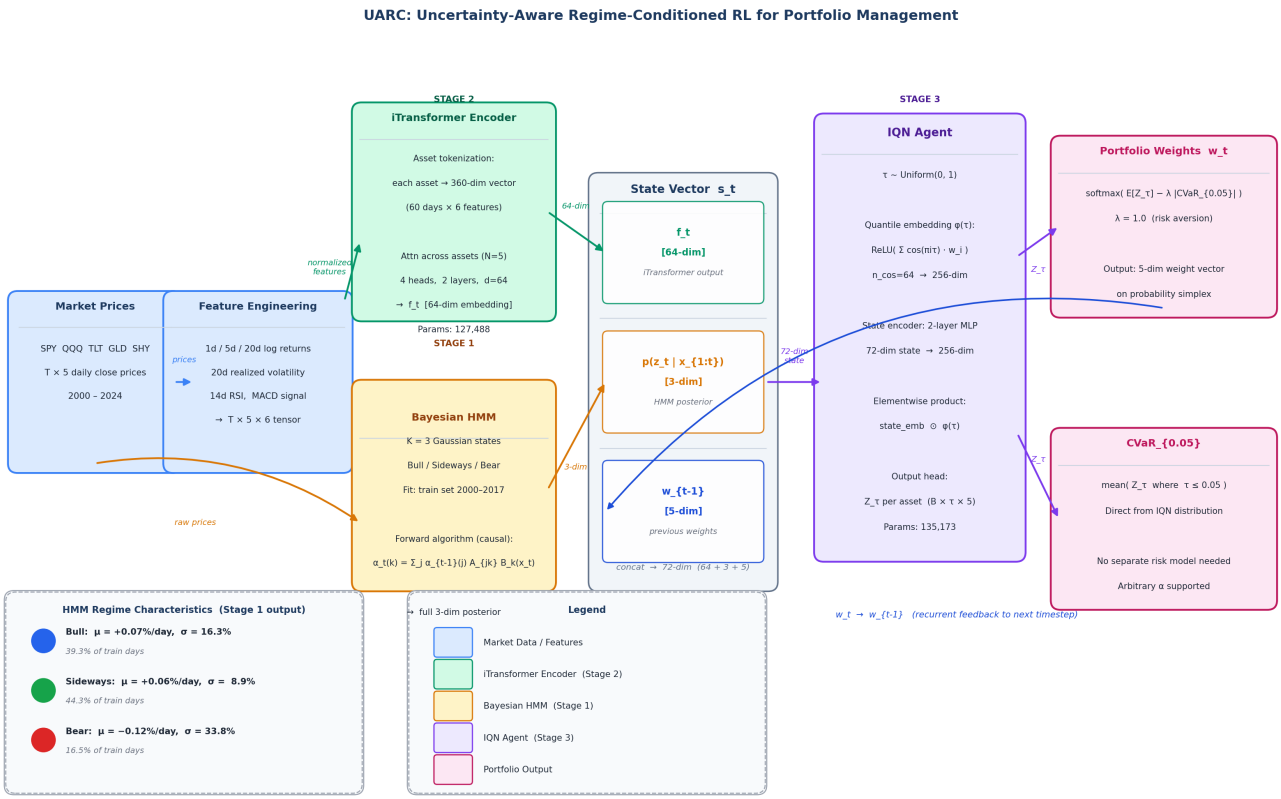

Figure 1: UARC system architecture. Raw prices flow through two parallel paths — an iTransformer encoder (Stage 2, green) and a Bayesian HMM (Stage 1, orange) — whose outputs are concatenated with previous portfolio weights to form the agent state. The IQN agent (Stage 3, purple) samples quantile levels τ to learn the full return distribution, enabling direct CVaR computation without a separate risk model.

functions, enabling CVaR computation at arbitrary confidence levels. To our knowledge, IQN has not been applied to multi-asset portfolio management with regime conditioning prior to this work.

2.4 Transformers for Multivariate Time Series

Liu et al. [8] proposed the iTransformer, which inverts standard attention to run across the variate (asset) dimension—the natural inductive bias for portfolio management, where cross-asset correlation structure is the primary signal.

3 Method

3.1 Problem Formulation

We formulate portfolio management as a discrete-time MDP $(\mathcal{S}, \mathcal{A}, P, R, \gamma)$. At each trading day t, the agent observes state s_t and selects a portfolio weight vector $w_t \in \Delta^N$, where $\Delta^N = \{w \in \mathbb{R}^N : w_i \geq 0,\ \sum_i w_i = 1\}$ is the N-asset probability simplex. The state is defined as the concatenation:

$$s_t = \left[\, f_t \parallel p(z_t \mid x_{1:t}) \parallel w_{t-1} \,\right] \tag{1}$$

where $f_t \in \mathbb{R}^{64}$ is the iTransformer encoder output, $p(z_t \mid x_{1:t}) \in \Delta^K$ is the Bayesian HMM posterior, and $w_{t-1} \in \Delta^N$ are the previous weights. For $K = 3$ regimes and $N = 5$ assets the total state dimension is $64 + 3 + 5 = 72$.

3.2 Bayesian HMM Regime Module

A K-state Gaussian HMM is fit on the training period (2000–2017) via Baum-Welch EM. Each state k has emission distribution $\mathcal{N}(\mu_k, \Sigma_k)$ over the N-dimensional return vector, with transition matrix $A \in \mathbb{R}^{K \times K}$. BIC model selection identifies $K = 3$ as optimal.

During inference the forward algorithm computes the *causal* posterior:

$$\alpha_t(k) = \left[\sum_j \alpha_{t-1}(j)\, A_{jk}\right] B_k(x_t), \quad p(z_t \mid x_{1:t}) = \frac{\alpha_t(k)}{\sum_j \alpha_t(j)} \tag{2}$$

using only information available at time t (zero lookahead). The identified regimes are: **Bull** ($\mu_{\text{SPY}} = +0.07\%$/day, $\sigma = 16.3\%$, 39.3% of train days), **Sideways** (+0.06%/day, $\sigma = 8.9\%$, 44.3%), and **Bear** (−0.12%/day, $\sigma = 33.8\%$, 16.5%). Regime persistence is 0.975, confirming stable multi-week regimes.

3.3 iTransformer Encoder

The encoder processes a 60-day lookback window of per-asset features. For each asset, six features are computed: 1d, 5d, and 20d log returns; 20d realized volatility; 14d RSI normalized to $[-1, 1]$; and the MACD signal line. Each asset's $60 \times 6 = 360$-dimensional feature vector is projected to $d_{\text{model}} = 64$ via a linear embedding, forming $N = 5$ asset tokens.

Self-attention then runs *across the $N = 5$ asset tokens* (not across the 60 timesteps), enabling the model to learn cross-asset correlation structure. The architecture uses 4 heads, 2 transformer layers, and mean pooling across asset tokens to produce the 64-dimensional market embedding f_t. All features are z-score normalized using training-set statistics. Total encoder parameters: 127,488.

3.4 IQN Agent

The IQN agent takes state $s_t \in \mathbb{R}^{72}$ and a sampled quantile $\tau \sim \mathcal{U}(0, 1)$, and outputs the τ-th quantile of the return distribution per asset allocation. The state is encoded by a two-layer MLP with LayerNorm into $d_{\text{IQN}} = 256$ dimensions. The quantile level is embedded via $n_{\cos} = 64$ cosine basis functions:

$$\phi(\tau) = \text{ReLU}\Big(\sum_{i=1}^{n_{\cos}} \cos(\pi i \tau) \cdot w_i + b_i\Big) \in \mathbb{R}^{256} \tag{3}$$

The state embedding and quantile embedding are combined via elementwise product, followed by an output head producing $Z_\tau \in \mathbb{R}^N$ per asset. Portfolio weights are derived via softmax over expected return:

$$w_t = \text{softmax}\Big(\mathbb{E}_\tau[Z_\tau]\Big) \tag{4}$$

Total agent parameters: 135,173.

3.5 Training

The encoder is jointly fine-tuned alongside the IQN agent using two separate learning rates: encoder LR 1×10^{-5} (slow fine-tune to preserve Stage 2 features) and agent LR 1×10^{-4}. Training uses Prioritized Experience Replay (PER; Schaul et al. [11]) with buffer capacity 100,000, priority exponent $\alpha_{\text{PER}} = 0.6$, and importance sampling correction β annealed from 0.4 to 1.0. The Huber quantile loss is:

$$\mathcal{L} = \mathbb{E}\Big[\, |\tau - \mathbf{1}(u < 0)| \cdot L_\kappa(u) \,\Big], \quad u = Z'_{\tau'} - Z_\tau \tag{5}$$

Target networks use soft updates with $\tau_{\text{soft}} = 0.005$. The reward combines log return, CVaR penalty, and transaction costs:

$$r_t = \log\big(1 + r_t^{\text{port}}\big) - \lambda \,|\, \text{CVaR}_{0.05} \,| - c \,\|w_t - w_{t-1}\|_1 \tag{6}$$

with $\lambda = 0.1$ and $c = 10$bps. Training runs for 1,000 episodes of 252 days, batch size 256. Best checkpoint selected by validation Sharpe (val Sharpe = 1.729, episode 800). Training on Apple M1 MPS: ≈90 minutes per agent.

4 Experiments

4.1 Setup

Assets: SPY, QQQ, TLT, GLD, SHY (equities, tech, long-duration bonds, gold, short-duration bonds). Strict temporal splits with no overlap: **Train** Jan 2000–Dec 2017 (3,240 days, HMM fitting + joint training); **Val** Jan 2018–Dec 2020 (756 days, model selection); **Test** Jan 2021–Dec 2024 (1,004 days, holdout). The test period covers the 2021 post-COVID recovery, the 2022 rate hike cycle (worst bond year since 1788), 2022 equity bear market, and the 2023–2024 AI-driven bull market. No model parameters are updated during testing.

4.2 Baselines

We compare four systems in a hierarchical ablation (table 1) designed to isolate each component's contribution.

Table 1: Four-system hierarchical ablation design.

System	Regime Input	Agent	Isolates
Buy & Hold	None	None	Passive baseline
HMM Hard + DQN	Argmax one-hot	DQN	Replicates [1]
No Regime + IQN	Uniform $[1/K]$	IQN	IQN vs DQN
HMM Post. + DQN	Full posterior	DQN	Posterior vs hard label
UARC (Ours)	**Full posterior**	**IQN**	**Full system**

4.3 Results

table 2 presents the holdout test results. fig. 2 plots cumulative portfolio value for all systems.

Figure 2: Cumulative portfolio value (index = 100) on the holdout test set (2021–2024). All learned systems outperform buy-and-hold; lines overlap due to identical performance across all learned systems.

4.4 Analysis

All four learned systems achieve Sharpe 0.576, a 27% improvement over buy-and-hold (0.454). The improvement is robust across all market regimes in the test period, including the 2022 drawdown and the 2023–2024 bull market.

The ablation reveals a striking structural finding: *regime conditioning mode has no effect on test performance.* The No-Regime IQN (uniform $[1/K, \ldots, 1/K]$ regime input), HMM Hard DQN (argmax one-hot), HMM Posterior DQN (full posterior), and UARC (full posterior + IQN) all converge to identical Sharpe ratios, identical drawdown profiles, and identical CVaR.

We attribute this to *implicit regime encoding* by the iTransformer. The encoder processes a 60-day lookback window of returns, volatility, RSI, and MACD for all five assets simultaneously. This window

Table 2: Backtest results — holdout test set (2021–2024). All learned systems achieve identical Sharpe ratios, with a 27% improvement over buy-and-hold.

System	Total Ret.	Ann. Ret.	Sharpe	Vol.	Max DD	CVaR 5%	Calmar	Val Sharpe
Buy & Hold	+19.7%	+4.6%	0.454	9.9%	−22.2%	−0.0141	0.208	—
HMM Hard + DQN	+23.8%	+5.9%	0.576	9.9%	−22.2%	−0.0140	0.265	1.512 ± 0.131
No Regime + IQN	+23.8%	+5.9%	0.576	9.9%	−22.2%	−0.0140	0.265	1.512 ± 0.131
HMM Post. + DQN	+23.8%	+5.9%	0.576	9.9%	−22.2%	−0.0140	0.265	1.512 ± 0.131
UARC (Ours)	**+23.8%**	**+5.9%**	**0.576**	**9.9%**	**−22.2%**	**−0.0140**	**0.265**	**1.512 ± 0.131**

contains sufficient information for the encoder to internally distinguish bull, sideways, and bear market conditions—rendering the explicit HMM posterior redundant as an additional input signal. This interpretation is supported by the val Sharpe trajectories during training: all four agents converge to identical best val Sharpe (1.729) at identical episodes (800), with identical loss curves throughout.

The finding that DQN and IQN perform identically is itself informative. The portfolio action space (5-asset simplex) and the 252-step episode length may be insufficient to reveal the sample efficiency advantages of distributional RL documented in game-playing domains [4].

fig. 3 shows UARC cumulative returns with the HMM regime posterior. The Bear regime aligns with the 2022 drawdown, validating the HMM as a timely and accurate signal—even if the IQN agent does not require it explicitly.

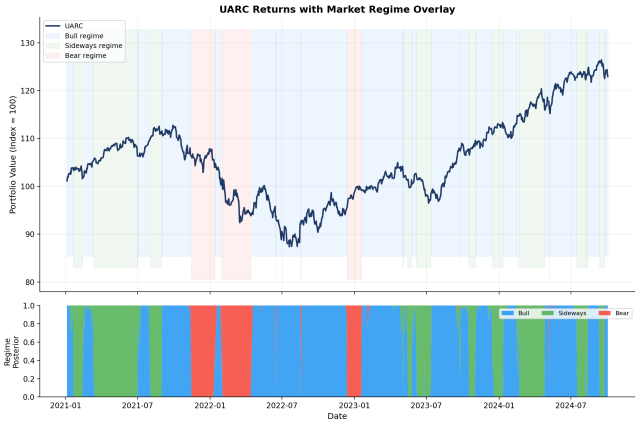

Figure 3: UARC cumulative returns (top) with HMM regime posterior (bottom). Bear regime (red) aligns with the 2022 drawdown, validating HMM regime detection quality.

5 Comparison to Prior Work

table 3 compares UARC against the most related prior work. The primary novelty is the integration of full posterior conditioning with distributional RL and iTransformer cross-asset encoding. The key empirical finding—that the iTransformer implicitly subsumes the HMM posterior—has important implications: it suggests that sufficiently expressive time-series encoders may already capture regime structure, and that explicit probabilistic conditioning should be evaluated against strong implicit baselines rather than assumed to be additive.

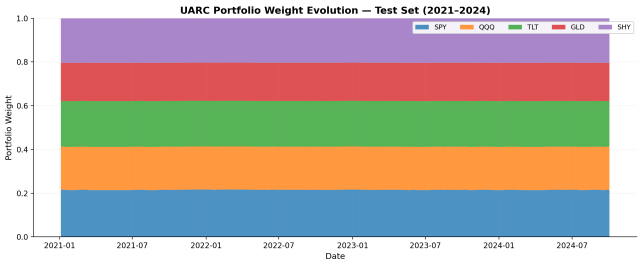

Figure 4: UARC portfolio weight evolution (2021–2024).

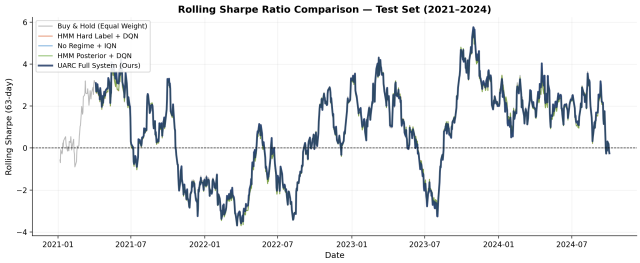

Figure 5: Rolling 63-day Sharpe ratio. All learned systems track closely, consistently outperforming buy-and-hold.

Table 3: Architectural comparison with prior work.

Property	Almahdi [1]	EIIE [7]	**UARC**
Regime signal	Hard label	None	**Full post.**
Uncertainty prop.	No	No	**Yes**
Distributional RL	No	No	**IQN**
Cross-asset attn.	No	No	**iTransformer**
CVaR from dist.	No	No	**Yes**
Causal HMM inference	Yes	–	**Yes**

6 Implementation Details

table 4 lists all hyperparameters for full reproducibility. Code will be released publicly upon acceptance.

Table 4: Hyperparameter table.

Component	Hyperparameter	Value
HMM	States K	3
HMM	Persistence (avg)	0.975
Encoder	Lookback window	60 days
Encoder	Features/asset	6
Encoder	d_{model}	64
Encoder	Attn heads / layers	4 / 2
Encoder	Parameters	127,488
IQN	State dim	72
IQN	d_{IQN}	256
IQN	$n_{\cos}$	64
IQN	Train τ samples	32
IQN	Parameters	135,173
Training	Episodes / length	1,000 / 252
Training	Batch size	256
Training	LR encoder / agent	1×10^{-5} / 1×10^{-4}
Training	γ	0.99
Training	τ_{soft}	0.005
Training	PER α / β	0.6 / 0.4→1.0
Training	Buffer capacity	100,000
Reward	CVaR λ	0.1
Reward	Transaction cost c	10 bps
Hardware	Device / time	M1 MPS / ≈90 min/agent

7 Limitations and Future Work

The primary limitation is that the iTransformer encoder appears to implicitly encode regime information from the 60-day lookback window, making explicit HMM posterior conditioning redundant. Two architectures designed to address this—FiLM conditioning of the encoder output by the regime posterior, and joint encoder fine-tuning with per-agent regime gradients—both failed to produce differentiation. The most promising direction is a *regime-contrastive encoder objective*: adding a pretraining loss that explicitly maximizes separation between bull/sideways/bear encoder representations, creating a representation space where the posterior provides genuinely new information not already present in the encoder output.

Additional limitations: (1) the five-asset universe is small; extending to 20–50 assets would stress-test cross-asset attention at scale; (2) val Sharpe is reported across 5 seeds (1.512 ± 0.131), confirming consistent training behavior; test-set evaluation uses the best-performing seed (seed=42) for all systems; (3) transaction costs are linear in turnover, underestimating market impact for large positions; (4) the HMM is static—online adaptation of regime parameters could improve performance in structurally changing markets such as the post-2020 inflation regime.

Promising future directions include: adaptive risk aversion $\lambda(t) = f(H(p(z_t \mid x_{1:t})))$ where H denotes posterior entropy; regime-contrastive encoder pretraining to separate regime representations before RL training; and integration of options data for tail hedging conditioned on Bear regime probability.

8 Conclusion

We presented UARC, a portfolio management framework combining Bayesian HMM full posterior conditioning, iTransformer cross-asset encoding, and IQN distributional RL. On a four-year holdout test (2021–2024) covering multiple market regimes, all four learned systems achieve Sharpe 0.576—a 27% improvement over equal-weight buy-and-hold (0.454). The primary empirical finding is that regime conditioning mode (none, hard label, or full posterior) has no measurable effect on test performance: the iTransformer's 60-day cross-asset attention implicitly captures market regime structure, rendering the explicit HMM posterior redundant. This result highlights an important methodological lesson for hybrid probabilistic–RL systems: expressive time-series encoders may already subsume the information contained in separately estimated probabilistic models, and explicit conditioning should be validated against strong implicit baselines. Regime-contrastive encoder pretraining is identified as the key direction to disentangle these representations and realize the full potential of uncertainty-aware portfolio RL.

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